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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination – APRIL 2021

ACH301 – ALLIED CHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. The Pi meson mass having
   a) 206 b) 270 c) 970 d) 306

2. The energy released in a fission reaction can be calculated from
   a) Binding energies of the nuclei b) Atomic number
   c) Mass defect d) Both (a) & (b)

3. The spectral range of IR spectra is
   a) 12500cm^{-1} b) 12500-25000cm^{-1} c)500-4000cm^{-1} d) 1-100cm^{-1}

4. Example for colligative property
   a) Depression in freezing point b) Osmotic pressure
   c) Lowering of vapors pressure d) All the above

5. The example of crystalline solids
   a) Sulphur b) Sugar c) Nacl d) All the above

6. The primitive cubic lattice which contains
   a) Corners of each unit cell b) Corners as well as in each cube
   c) Centers of each cube d) All the above

7. In amperometric titration
   a) Voltage is kept constant b) Current is kept constant
   c) Voltage is measured d) b&c

8. The indicator electrode is used in
   a) Complexo metric reaction b) Precipitation reaction
   c) Conductometric titration d) Amperometric titration

9. The perovskite structure is an example of
   a) LTSC b) HTSC c) Both (a)& (b) d) None of the above

10. The standard electrode potential of lead in hydrochloric acid is
    a) Positive b) Negative c) Either positive (or) negative d) None of the above

II. Fill up the blanks

11. All the radioactive elements having their atomic number is above _____.

12. Spectroscopy mainly deals with_____.

13. The miller indices of a crystals are _____ to the intercepts of that face on the various axes.

14. The titration of silver nitrate and potassium chloride is an example of _____titration.

15. Law of normality is _____.

III. Match the following

16. Nuclear force	- Bravais lattice
17. Elevation of boils point	- Fe^{2+}/Ce^{4+}
18. FCC	- LTSC
19. Redox titration	- Attractive force
20. Meissner effect	- Colligative property
	- Super conductivity

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. What are magic numbers?
22. State nuclear fission reaction?
23. Write note on: Nuclear reaction.
24. Distingish between absorption and emission spectrum.
25. Define relative lowering of vapour pressure.
26. What is lattice energy of an ionic crystal?
27. What is mean by space lattice?
28. Define the term titration curve.
29. What are super conductors? Give any two characters.
30. Write the electrode reactions for Hydrogen and Chlorine electrodes.
31. Write note on: Quinhydrone electrode.
32. What is corrosion of a metal?

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Write an essay of nuclear reactions. **(5)**
    b) Define the term: i) Alpha particle ii) β-particle **(2)**
34. a) Write the differences between nuclear fission and nuclear fusion. **(4)**
    b) Write a note on the applications of radioactivity. **(3)**
35. a) Define elevation in boiling points. **(2)**
    b) Explain about i) allowed & forbidden transitions **(3+2)**
    ii) Selection rules for IR radiation
36. Derive Bragg equation in X-ray crystallography. **(7)**
37. Compare electron and neutron diffractions. **(7)**
38. a) Explain about amperometry titration with its application. **(3.5)**
    b) Write a note on complex metric titration. **(3.5)**
39. a) Derive an expression for Nernst equation. **(4)**
    b) How corrosion can be controlled of a metal? **(3)**

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